Author, year, country	Phantom type and materials	Robustness evaluation	Stability threshold	Parameters evaluated	MRI scanning details (vendor, model, sequence, field strength)	Segmentation and radiomics software, type and number of extracted features	Main results
Lee et al., 2021, USA [1]	Physical, acrylic filled with water and 20 embedded inserts[1]	Repeatability (test-retest) and reproducibility	ICC ≥ 0.9 COV<10%	MRI acquisition parameters, vendor type (Siemens and Philips). Impact of image preprocessing[2]	1.5T Siemens and Philips, 3D T1WI GE and T2WI FSE sequences	Semiautomated, IBEX, 76 first and second order texture	Repeatability was higher in phantoms than in volunteers. T1WI showed lower variability than T2WI in the absence of preprocessing. Preprocessing had no significant effect on feature robustness
Rai et. al., 2022, Australia [2]	Physical, 3D printed with different shapes and patterns using MRI visible polymer resin	Longitudinal repeatability[3] and reproducibility	ICC ≥ 0.8 COV ≤ 5%	Image noise and resolutions	3T MAGNETOM Skyra, Siemens, T2WI TSE sequence	Semiautomated, PyRadiomics, 83 shape, first and second order texture	First order features had higher stability to changes in image resolution and noise than second order texture features. In the temporal stability test, approximately 49% of features had COV ≤ 10%
Sghedoni et al., 2025, Italy [3]	Physical, abdomen-shaped elliptical PMMA phantom[4] filled with silicon oil and 4-cylinder	Repeatability (test retest) and reproducibility	COV, ICC, no threshold	(among 19 scanners with different imaging parameters). Impact of image	19 scanners: 14 1.5T and 5 3T for three vendors: Siemens, GE, Philips, each with	Semiautomated, PyRadiomics, 107, shape, first and second order texture	Feature reproducibility improved after image preprocessing, except for GLRLM features. The repeatability was < 7% for all feature

	tubes filled with plastic spheres of different sizes			preprocessing (bias field correction, denoising, intensity normalization, and longitudinal stability [5]	different models [6]. T2WI acquired.		families and remained unaffected by the applied image pre-processing. In general pre-processing steps applied enhanced inter-scanner reproducibility, while having no impact on intra-scanner repeatability.
Bologna et al., 2022, Italy [4]	Organic cucumber placed in plastic container [7]	Repeatability (test retest) and reproducibility	ICC > 0.75 and mean absolute COV < 0.3	Imaging acquisition [8]. Impact of image preprocessing addressed.	Siemens Magnetom Avanto 1.5T, T1WI, T2WI, and DWI (0 and 1000 s/mm^2 with monoexponentially fitting for ADC calculation)	Manual, 4 ROI (heterogenous and homogenous parts of cucumber) PyRadiomics, 107, shape, first and second order texture	Preprocessing increases features stability to both test–retest and variation of the image acquisition parameters for all the types of analysed MRI images, except for slice thickness. T2WI-based features have higher ICCs and lower CVs compared to those in ADC and T1WI.
Cattell et al., 2019, USA [5]	Organic pineapple, banana, orange and kiwi placed on Styrofoam box	Reproducibility	ICC > 0.9	SNR, acquisition voxel size and ROI transformation.	3T Siemens (Biograph mMR) with a T2WI TSE sequence	Manual, in-house software, 56, shape, first and second order texture	Dilation resulted in poorer robustness when compared to erosion of ROI. Compared to texture

				Impact of normalization in stability is addressed[9]			features, first order features are high robust to added noise. Both methods of intensity normalization improve the robustness of texture features.
Shur et al., 2021, UK [6]	Water and tissue phantom	Repeatability (test retest) and reproducibility	COV< 5%, CCC>0.99	Noise and image resolution and phantom repositioning	1.5-T Siemens Magnetom Aera scanner, T1WI and T2WI	Manual, in-house software, 46 first and second order texture	Out of the 46 extracted features, 11 robust features (COV<5%) with image noise and resolution. For test retest, 38 and 36 robust features for T1WI and TWI, respectively
Sun et al., 2022, China, [7]	Ten jellies, five fruits/vegetables phantom[10], and a dynamic phantom (CIRS 008Z)	Repeatability (test retest by) and reproducibility	DSC, CCC>0.9	Interobserver variance, slice thickness, radiation status, motion, and phantom repositioning[11] (multivariate analysis)	1.5T integrated MR-LINAC, 3D T2WI	Manual, PyRadiomics, 1409, shape, first, second and transformed features[12]	Shape features showed the highest robustness in test-retest analysis. 40.0% of features showed excellent robustness across the different thicknesses. Shape and first order features were the highest robust in interobserver analysis.

							Shape and NGTDM features were both the most stable features with the beam both on and off, regardless of the beam position. More stable features when beam off. Only 4 features remained stable for amplitudes of 5–15 mm. 25 (2.4%) features with excellent robustness under five influences. Shape features are the most stable and reproducible features in all tests except the effect of motion.
Yu et al., 2024, China [8]	ACR phantom[13]	3 repeatability tests and 2 reproducibility tests. Longitudinal repeatability, test retest and reproducibility	ICC>0.9 CV<10 %	Two scanners, inter and intra observer (2 weeks) agreement, longitudinal stability (30 days), phantom repositioning	Integrated 1.5T MRI with 7MV flattening-filter-free electron LINAC system, and 3D T1WI (FFE), T2WI (TSE), and FLAIR images	Manual, PyRadiomics, 94 first and second order texture	FLAIR showed the highest number of robust features compared to T2WI and T1WI for repeatable and reproducible assessments.

Baeßler et al., 2019, Germany [9]	Biological phantom (onions, limes, kiwifruits, and apples placed on Styrofoam box)	Repeatability (test-retest) and reproducibility	CCC ≥0.9, DR ≥ 0.9, ICC≥0.75	Phantom repositioning, image resolution (matrix size), inter and intra observer (2 weeks) agreement	Ingenia, Philips, 3T, T1WI, T2WI, and FLAIR	Semiautomated, LIFEX, 46, shape, first and second order texture	FLAIR showed the highest number of robust features compared to T2WI and T1WI for repeatable and reproducibility assessments. 15 of 45 (33%) features showed excellent robustness and reproducibility across all sequences.
Wichtmann et al., 2023, Germany [10]	Biological phantom (onions, limes, kiwifruits, and apples placed on polystyrene foam box)	Features value and its repeatability (test-retest)	ANOVA, CCC ≥0.9, DR ≥ 0.9	Phantom repositioning, intraobserver (2 weeks) agreement, spatial resampling, intensity discretization, and intensity rescaling [14]	Ingenia, Philips, 3T, T1WI, T2WI, and FLAIR with low and high resolution (matrix size)	Semiautomated, LIFEX, 45, shape, first and second order texture	Intensity discretization impacted feature values most, whereas intensity rescaling had no significant effect on feature values except those from T1WI. In test retest, FLAIR sequence in high-resolution overall yielded the highest number of repeatable features in comparison with T1WI and T2WI
Bernatz et al., 2021, Germany [11]	Biological phantom (onions, limes, kiwifruits, and apples	Repeatability (test retest) and reproducibility	CCC ≥0.9, DR ≥ 0.9, ICC≥0.75, Gini score > 107 successes for	Inter and intra-observer agreement, phantom repositioning	Magnetom, Prisma, Siemens, 3T, T2WI HASTE, T2WI TSE, T2WI	Semiautomated,106 PyRadiomics, shape, first and second order texture	T2 map had the highest feature robustness, high intra and inter-observer reproducibility and

	placed on Styrofoam box)		discriminative performance of features (to differentiate between fruits)		FLAIR, T2 map and T1WI TSE		most promising discriminative power
Rai et al., 2020, Australia, [12]	3D printed phantom (different shapes and patterns using RGD-525 MRI visible resin)	Longitudinal repeatability (over 12 months) and reproducibility	COV≤ 5%, ICC>0.8	Inter-scanner (8 for 3 vendors with 6 model types)	1.5 and 3T, Siemens (Skyra, Vida, Aera), Philips (Ingenia), GE (Optima360, Signa HDxt), T2WI TSE sequence	Semiautomated, in-house developed, 69 shape, first and second order texture	First-order statistics were the most robust in stability across the eight scanners and no shape features had high stability features across the eight scanners.
Jang et al., 2020, USA, [13]	Biological phantom (onions, limes, kiwifruits, and apples)	Repeatability (test-retest) and reproducibility	ICC≥0.80	Phantom repositioning	3T Siemens (Vida), cine balanced SSFP, T1WI, T2WI, T1 and T2 mapping	Manual, PyRadiomics, 1023, first, second and transformed[15] texture features	Within and between sessions, the First order and GLCM feature families demonstrated the highest reproducibility, with no single image filter consistently producing more reproducible features than others.
Bianchini et al., 2020, Italy, [14]	In-house developed phantom[16] (background filled with $MnCl_2$ and 4 inserts	Repeatability (longitudinal and test-retest scans)	ICC>0.9	Phantom repositioning and temporal stability (10 scans same position)	1.5T, Optima MR450W, GE, T1WI, T2WI	Manual, PyRadiomics,1034, shape, first second order and transformed [17]texture	Out of 1,034 extracted features, 33% were found to be repeatable on T1-weighted images (T1WI), and 25% on T2-

	mimic relaxation times and texture of female pelvic tumours)						weighted images (T2WI).
Mi et al., 2020, China, [15]	Biological phantom (kiwi)[18] and in vivo (9 healthy cervix volunteers)	Repeatability and reproducibility	CV < 0.1 and QCD < 10	Inter scanner (3 scanners) and 4 imaging acquisition parameters (TR, TE, slice thickness, and acquisition matrix). Also, impact of intensity normalization on robustness	3T, Siemens (Skyra), GE (Signa HDxt) and Philips (Ingenia), T2WI without FS,	Manual, A.K. radiomics software, 396, shape, first and second order texture	In the kiwi phantom, 51.5% of the extracted features were reproducible across the three MRI scanners. Furthermore, both the reproducibility and the discriminative power of robust features—specifically for distinguishing between the junctional zone and the outer muscular layer of the healthy cervix—were enhanced through intensity normalization.
Veres et al., 2022, Hungary, [16]	QR code and Hilber cube-based 3D printed phantom using PLA filled with 0.01 mM $NiCL_2$ solution	Repeatability (3 scans) and reproducibility	COV ≤ 10%, RPD, and ICC	Multivariate analysis of spatial resolution, magnetic field strength, intensity	3T Philips Achieva and a 1.5T Siemens Magnetom Essenza. 3D T1 MPRAGE protocol for T1WI,	Semiautomated, in-house developed, 45, first and second order texture	Repeatability and reliability of radiomic features were higher in data acquired at 1.5 T compared to 3 T, even when high spatial resolution was used.

	and biological phantom (kiwis, tomatoes, and onions)			discretization (FBN and FBS), number of coil's channel and biological phantom orientation	and 3D Brain VIEW and 3D T2 SPACE for T2WI		Additionally, good agreement was observed between results obtained from 3D-printed phantoms and biological phantoms. The choice of discretization method did not significantly impact repeatability measures. However, a larger coefficient of variation (COV) was observed when the phantom was rotated, compared to when it was positioned in its original orientation.
Michalet et al., 2024, France, [17]	QA phantom (ACR phantom) and in vivo (pancreatic cancer)	Longitudinal repeatability	COV ≤ 5% and ICC ≥ 0.90	Repeated scan (55 scans over 3 weeks for phantom and MR image simulation and first fraction MR image in in vivo investigation)	0.35T, integrated MRI-Linac, balanced steady-state gradient echo sequence (True FISP)	Semiautomated, PyRadiomics, 107 shape, first and second order texture	Out of the total features, 44 (47%) demonstrated high repeatability between sessions in the phantom, with COV ≤ 5%. In in vivo investigations, 28 features (26%) showed high reproducibility

							across repeated scans.
Ericsson-Szecsenyi et al., 2022, Sweden, [18]	Magphan RT phantom and View Ray Daily QA phantom[19] and in vivo (pancreatic cancer)	Longitudinal repeatability and reproducibility	COV < 5%	Repeated scans (11 scans over 13 months and 11 scans over 11 days)	0.35T, integrated MRI-Linac, True Fast Imaging with Steady-State Free Precession (TRUFI) pulse sequence	Manual, in-house developed, 1087, shape, first, second and high order texture[20]	130 robust features across the two phantom and in vivo (both kidneys and liver) investigation.
Wong et al., 2020, China, [19]	QA phantom (ACR phantom)[21]	Longitudinal repeatability and reproducibility	COV < 5%, ICC > 0.75 and ratio of the interobserver difference to the interobserver mean δ < 5%.	Repeated scans (30 days) and interobserver agreement	Two 1.5T MR simulators, Magnetom Aera Siemens and Ingenia MR-RT, Philips, 3D T1WI TSE sequence with SPACE and Brain VIEW protocol type, respectively.	Manual, PyRadiomics, 101, shape, first and second order texture	Across both scanners, shape features showed the highest longitudinal repeatability, while first-order and texture features had lower interobserver variation and better agreement. Texture features also demonstrated significant longitudinal repeatability between scanners.
Ammari et al., 2021, France, [20]	In-house developed homogenous and heterogenous phantoms[22] and in vivo (10	Reproducibility	Student's t-test	Magnetic field strength, matrix size and FOV	Two GE scanners (an Optima 1.5T and Discovery 3T), 3D SPGR sequence	Manual, LIFEX, 34, first and second order texture	Radiomic features extracted at 1.5 T should not be used interchangeably with those from 3 T for texture analysis, as they are influenced by

	healthy volunteers)						field strength, pixel size, field of view (FOV), and matrix size.
Bologna et al., 2019, Italy, [21]	Virtual phantom	Reproducibility	ICC > 0.75	TR (350 – 650 ms) and TE (5 – 15 ms), and effect of three intensity normalization on feature stability (z-score, histogram matching, and linear standardization)	T1WI, SE sequence, using virtual software for MRI scan simulation	Manual, PyRadiomics, 58, first and second order texture	Most radiomic features (76%) demonstrated good reliability (ICC > 0.75). Intensity standardization enhanced the reliability of first-order features, while histogram matching significantly reduced that of GLCM features.
Mayerhoefer et al., 2009, [22]	In-house developed phantom (polystyrene spheres and agar gel)[23]	Reproducibility and pattern discrimination ability of features	LDA and KNN	Number of acquisitions, TR, TE, and sampling bandwidth at different matrix sizes.	3T, T2WI, multi-echo SE sequence	Manual, MAZDA, second and high order texture features	Texture features across all categories become increasingly sensitive to acquisition parameter variations as spatial resolution increases. However, when spatial resolution is sufficiently high, variations in NA, TR, TE, and SBW have minimal impact on pattern discrimination

Xue et al., 2021, China, [23]	Anthropomorphic phantom (CIRS triple-modality 3D Abdominal Phantom)[24]	Accuracy, repeatability (5 repetitive scans) and reproducibility	ICC >0.90, COV <5%, and percentage deviation (from reference non-accelerated reconstructed MR image) <5%	Two acceleration (SENSE and compressed SENSE) image reconstruction algorithms with different acceleration factor, direction, and denoising level	MRI simulator, Ingenia MR-RT, Philips, 1.5T, 3D T2WI TSE sequence,	Semiautomated, PyRadiomics, 93 first and second order texture	Radiomics features demonstrated greater accuracy and robustness under SENSE reconstruction compared to compressed SENSE, yielding more reliable features. Reconstruction method had no effect on feature repeatability. First-order features exhibited the highest accuracy across all acceleration factors for both algorithms.
Saint Martin et al., 2020, France, [24]	Anthropomorphic phantom (two modality breast biopsy and sonographic phantoms, CIRS reference 073) [25]	Reproducibility across 30 ROI of gel mimicking normal tissue	COV and Krustal Wallis test	Two scanners and three dual breast coils (8,16,18 channels coils). Bias field correction, intensity normalization (z-score and histogram	magnet, Optima MR450w, GE (8 channel coil) and MAGNETOM Aera, Siemens (16 and18 channel coil), T2WI_FS_TSE, T1WI and T1WI-DCE (SPGE sequence)	Manual, LIFEX, 42, first and second order texture	ComBat harmonization reduced the proportion of radiomic features significantly differing between the three coils from 87% (post–bias field correction and MR normalization) to 3% in the gel, while maintaining or enhancing lesion

				matching) and harmonization.			classification performance in the phantoms.
Bianchini et al., 2021, Italy, [25]	In-house developed phantom (pelvic shaped)[26]	Repeatability (test retest and longitudinal) and reproducibility	ICC >0.9, CCC>0.9	Phantom repositioning, inter scanner (3 scanners, from 2 vendors, 2 magnetic field strength), TR, TE	1.5T Optima MR450W, GE, and 1.5T Ingenia, Philips, and 3T Ingenia, Philips, T2WI	Manual, PyRadiomics, 944, shape, first, second, and high order texture[27]	Without repositioning, 92.1–96.4% of features were repeatable at 1.5T, dropping to 79.9% at 3T. Repeatability decreased further with repositioning and with increasing TE and TR.
Mitchell-Hay et al., 2024, Scotland, [26]	In-house developed phantom (9 tubes of agarose gel embedded within polyethylene beads)[28]	Repeatability (longitudinal) and reproducibility	ICC >0.9, COV <5%	Repeated scans 10 times over 4 months. Also, a range of TR and TE	Philips 3T Achieva dStream, T2WI	Manual, PyRadiomics, 1596, first, second and transformed texture [29]	First order features had the highest proportion of excellent repeatability across both TE and TR values. Wavelet filters yielded the most stable features across TE and TR variations, while exponential and square root filters showed greater sensitivity to parameter changes.
Dreher et al., 2020, Germany,	Biological phantom (onions, mandarins, kiwifruits, and	Repeatability (test retest) and reproducibility	ICC>0.90	Phantom repositioning, intra (after 2 weeks same	1.5 T, Magnetom AERA, Siemens, EPI_ DWI sequence	Semiautomated, LIFEX, 46, shape, first and second order texture	DWI yielded robust radiomic features, with those from ADC showing slightly lower

[27]	apples fixed on a Styrofoam-based plate)			rater) and inter observer variability and spatial resolution	(b=0,500,1,000 s/mm^2) and ADC calculated (mono-exponential) with low (3×3 mm^2) and high (2×2 mm^2) image resolutions.		robustness than features from raw DWI (b = 500, 1000 s/mm^2). No significant differences were observed across spatial resolutions. Inter- and intra-observer reliability remained consistently high across all sequences.
Hajianfar et al., 2024, Switzerland, [28]	In-house developed phantom (18 gel-filled tubes (25 mm) and one 20-mm tube filled with 0.25 mM Gd DTPA)	Reproducibility and repeatability and the impact of ComBat harmonization on feature robustness	ICC >0.90 and Kruskal-Wallis	Inter-scanner variability (four scanners), longitudinal scans (three scans at 1-hour intervals), and acquisition and preprocessing parameters[30]	2 GE 1.5T, Siemens 1.5T, and GE 3.0T, 3D FSPGR (Multiple FA, 2 – 30º), FSE_IR (Multiple IR, 50 – 3000 ms),	Manual, PyRadiomics, 92, first, second and high order texture	Variations in scanner models, inversion recovery times (IRs), flip angles (FAs), and repeated scans significantly influence radiomic features. ComBat harmonization effectively improves feature stability across these sources of variability.
Simpson et al., 2020, USA, [29]	In-house developed phantom (four different texture inserts)[31]	Reproducibility	COV < 10%,	Three intensity quantization methods with different	An integrated MRI-Linac, 0.35T, balanced SSFP	Manual, in-house developed, 39 second order texture	GLCM-based entropy and homogeneity, along with key GLRLM features, showed COV < 10% across all

				number of grey level bins[32]			combinations of quantization methods and number of bins for all 4 inserts. No GLSZM or NGTDM features met this threshold consistently. Histogram equalization to 64 grey levels yielded the highest number of low-variability texture features across all 4 phantom inserts.
Yuan et al., 2021, China, [30]	Anthropomorphic phantom (CIRS triple-modality 3D Abdominal Phantom)[33]	Reproducibility	ICC ≥ 0.90, percentage deviation < 5%	Nine imaging acquisition parameters[34]	An integrated MRI-Linac, Ingenia MR-RT, Philips, 1.5T, 3D-T2WI-TSE sequence	Semiautomated, PyRadiomics, 1395, first, second and high order[35] texture	First-order and GLCM features were generally more robust to imaging parameters in original images. In contrast, significant differences in radiomic feature values were observed between original and transformed images, indicating variable robustness to imaging changes.
Li et al., 2021, France,	Same phantom for reference 20.	Reproducibility	Friedman test or Wilcoxon test	Magnetic field strength and image	Optima MR450w 1.5T and Discovery	Manual, PyRadiomics, 92,	ComBat (feature harmonization level) effectively removes

[31]	In-house developed homogenous and heterogenous phantoms[36] and in vivo (6 healthy volunteers and MR images of brain tumour patients)			resolution (FOV and matrix size). Also, preprocessing (N4 bias field correction and image resampling) as well as harmonization (6 intensity image normalization methods or features using ComBat) is studied.	MR750w 3T, both from GE, T1WI	first and second order texture	scanner effects in brain MRI radiomics. None of the applied image intensity normalization methods eliminate scanner-related variability. Image resampling does not eliminate scanner effects, with mean intensity gaps between 1.5T and 3T MRI remaining.

Reference

1. Lee, J., *Radiomics feature robustness as measured using an MRI phantom Sci*. 2021, Rep.
2. Rai, R., et al., *Repeatability and reproducibility of magnetic resonance imaging-based radiomic features in rectal cancer.* Journal of Medical Imaging, 2022. **9**(4): p. 044005-044005.
3. Sghedoni, R., et al., *Stability of radiomic features in magnetic resonance imaging of the female pelvis: A multicentre phantom study.* Physica Medica, 2025. **130**: p. 104895.
4. Bologna, M., et al., *Repeatability and reproducibility of MRI‐radiomic features: A phantom experiment on a 1.5 T scanner.* Medical Physics, 2023. **50**(2): p. 750-762.
5. Cattell, R., S. Chen, and C. Huang, *Robustness of radiomic features in magnetic resonance imaging: review and a phantom study.* Visual computing for industry, biomedicine, and art, 2019. **2**: p. 1-16.
6. Shur, J., et al., *MRI texture feature repeatability and image acquisition factor robustness, a phantom study and in silico study.* European Radiology Experimental, 2021. **5**: p. 1-11.
7. Sun, M., et al., *Robustness and reproducibility of radiomics in T2 weighted images from magnetic resonance image guided linear accelerator in a phantom study.* Physica Medica, 2022. **96**: p. 130-139.
8. Yu, H., et al., *Quantifying the reproducibility and longitudinal repeatability of radiomics features in magnetic resonance Image-Guide accelerator Imaging: A phantom study.* European Journal of Radiology, 2024. **181**: p. 111735.
9. Baeßler, B., K. Weiss, and D.P. Dos Santos, *Robustness and reproducibility of radiomics in magnetic resonance imaging: a phantom study.* Investigative radiology, 2019. **54**(4): p. 221-228.
10. Wichtmann, B.D., et al., *Influence of image processing on radiomic features from magnetic resonance imaging.* Investigative Radiology, 2023. **58**(3): p. 199-208.
11. Bernatz, S., et al., *Impact of rescanning and repositioning on radiomic features employing a multi-object phantom in magnetic resonance imaging.* Scientific reports, 2021. **11**(1): p. 14248.
12. Rai, R., et al., *Multicenter evaluation of MRI‐based radiomic features: A phantom study.* Medical physics, 2020. **47**(7): p. 3054-3063.
13. Jang, J., et al., *Reproducibility of segmentation-based myocardial radiomic features with cardiac MRI.* Radiology: Cardiothoracic Imaging, 2020. **2**(3): p. e190216.
14. Bianchini, L., et al., *PETER PHAN: An MRI phantom for the optimisation of radiomic studies of the female pelvis.* Physica Medica, 2020. **71**: p. 71-81.
15. Mi, H., et al., *Impact of different scanners and acquisition parameters on robustness of MR radiomics features based on women's cervix.* Scientific reports, 2020. **10**(1): p. 20407.

16. Veres, G., et al., *Phantom study on the robustness of MR radiomics features: Comparing the applicability of 3D printed and biological phantoms*. Diagnostics, 2022. **12**(9): p. 2196.
17. Michalet, M., et al., *Robustness of radiomics features on 0.35 T magnetic resonance imaging for magnetic resonance-guided radiotherapy.* Physics and Imaging in Radiation Oncology, 2024. **31**: p. 100613.
18. Ericsson-Szecsenyi, R., et al., *Robustness assessment of images from a 0.35 T scanner of an integrated MRI-Linac: Characterization of radiomics features in phantom and patient data*. Technology in Cancer Research & Treatment, 2022. **21**: p. 15330338221099113.
19. Wong, O.L., et al., *Longitudinal acquisition repeatability of MRI radiomics features: an ACR MRI phantom study on two MRI scanners using a 3D T1W TSE sequence.* Medical Physics, 2021. **48**(3): p. 1239-1249.
20. Ammari, S., et al., *Influence of magnetic field strength on magnetic resonance imaging radiomics features in brain imaging, an in vitro and in vivo study.* Frontiers in Oncology, 2021. **10**: p. 541663.
21. Bologna, M., V.D. Corino, and L.T. Mainardi. *Assessment of the effect of intensity standardization on the reliability of T1-weighted MRI radiomic features: experiment on a virtual phantom*. in *2019 41st Annual International Conference of the IEEE Engineering in Medicine and Biology Society (EMBC)*. 2019. IEEE.
22. Mayerhoefer, M.E., et al., *Effects of MRI acquisition parameter variations and protocol heterogeneity on the results of texture analysis and pattern discrimination: an application-oriented study.* Medical physics, 2009. **36**(4): p. 1236-1243.
23. Xue, C., et al., *Reliability of radiomics features due to image reconstruction using a standardized T2-weighted pulse sequence for MR-guided radiotherapy: An anthropomorphic phantom study.* Magnetic Resonance in Medicine, 2021. **85**(6): p. 3434-3446.
24. Saint Martin, M.-J., et al., *A radiomics pipeline dedicated to Breast MRI: validation on a multi-scanner phantom study.* Magnetic Resonance Materials in Physics, Biology and Medicine, 2021. **34**: p. 355-366.
25. Bianchini, L., et al., *A multicenter study on radiomic features from T2-weighted images of a customized MR pelvic phantom setting the basis for robust radiomic models in clinics*. Magnetic resonance in medicine, 2021. **85**(3): p. 1713-1726.
26. Mitchell-Hay, R., et al., *Phantom study investigating the repeatability of radiomic features with alteration of image acquisition parameters in magnetic resonance imaging*. Journal of Medical Imaging and Radiation Sciences, 2024. **55**(1): p. 19-28.
27. Dreher, C., et al., *Radiomics in diffusion data: a test–retest, inter-and intra-reader DWI phantom study.* Clinical radiology, 2020. **75**(10): p. 798. e13-798. e22.
28. Hajianfar, G., et al., *Impact of harmonization on the reproducibility of MRI radiomic features when using different scanners, acquisition parameters, and image pre-processing techniques: a phantom study.* Medical & biological engineering & computing, 2024. **62**(8): p. 2319-2332.

29. Simpson, G., et al., *Impact of quantization algorithm and number of gray level intensities on variability and repeatability of low field strength magnetic resonance image-based radiomics texture features.* Physica medica, 2020. **80**: p. 209-220.
30. Yuan, J., et al., *Quantitative assessment of acquisition imaging parameters on MRI radiomics features: a prospective anthropomorphic phantom study using a 3D-T2W-TSE sequence for MR-guided-radiotherapy.* Quantitative imaging in medicine and surgery, 2021. **11**(5): p. 1870.
31. Li, Y., et al., *Impact of preprocessing and harmonization methods on the removal of scanner effects in brain MRI radiomic features.* Cancers, 2021. **13**(12): p. 3000.

Footnotes:

---

[1] Composed of materials at varying concentrations, including Styrofoam balls, petroleum jelly, Polyteck gel, clear ballistic gel, liquid plastic, Superflab, and others

[2] Number of excitations, slice thickness, phasing steps, and field of view as imaging parameters. Intensity normalization and Butterworth filtering as preprocessing

[3] Temporal stability against 13 scans over 4 months. Image gaussian noise of 2, 5, 20, and 20 SD and image resolution of 0.5, 0.8, 1.0 and 1.2 mm^2.

[4] With 30 cm, 24 cm, and 20 cm major axis, minor axis, and height, respectively. Four inserts filled with polystyrene spheres of different diameters (1–8 mm) and 0.1 % and 0.5 % water solution of agar, each with 6 cm inner diameter and T1 ≃ 1200 ms and T2 ≃ 120 ms at 1.5T

[5] N4 algorithm for bias field, spatial-adaptive Non-Local Means (SANLM) denoising filter, and histogram matching algorithm for normalization. The impact of suboptimal coil (affect SNR) on radiomics also assessed.

[6] For Siemens: Aera, Biograph_mMR, SymphonyTim, Avanto Fit, Skyra, Skyra Fit. For Philips: Achieva dStream, Ingenia, Achieva. For GE: SIGNA Explorer, Optima.

[7] 10 cm in diameter and 20 cm in height and polyester used for fixation.

[8] Against 4 parameters including TR, TE, slice thickness, and pixel spacing. Preprocessing including voxel size resampling (B-spline interpolation) (1mm for T1WI and T2WI, while 3 mm for ADC map) and intensity normalization (for T1WI and T2WI using z-score).

[9] Ten different noise realizations were performed numerically for two SNR level (45 and 75), mean ± 3SD and zero to maximum as intensity normalization, dilation and erosion by 1 pixel for ROI delineation, and different in-plane resolutions of 0.47, 0.50, 0.56 and 0.67 mm

[10] water phantom is a thorax-shaped container filled with a solution of 0.4 mM $Mncl_2$ to simulate the relaxation time T2 of muscle surrounding thoracic lesion, inside it jellies and fruits/vegetables inserted. Dynamic phantom body has a width of 32 cm, a height of 18 cm, and a length of 25.6 cm. It has a jellies and fruits/vegetables glued in the bottle one after another, moving in amplitudes of 0, 5, 8, and 15 mm.

[11] Image segmented twice by same oncologist, three slice thicknesses (1.2, 2.4, 4.8mm) investigated

[12] Using original, wavelet, square, logarithmic, exponential and gradient images

[13] characterized by hollow cylinders of acrylic plastic sealed at both ends, filled with a solution containing 10 mM $NiCl_2$ and 75 mM NaCl. with 190 mm and 148 mm internal diameter and length, respectively.

[14] resampled to isotropic voxels ranging from 0.75 to 10 mm, intensity discretized to fixed number of bins either 8, 16, 32, or 64 Nb, and intensity rescaled to either mean value ± 3 SD or minimum and maximum values.

[15] Wavelet, Square, Square Root, Logarithm, Exponential, Gradient, and local binary pattern 2D filters

[16] NEMA IEC pelvis-shaped container with background compartment filled with $MnCl_2$ to reproduce the relevant (T_1 or T_2) relaxation time of muscle surrounding pelvic malignancies. Four-cylinder Inserts (48 mm diameter, 72 mm height) filled with mixture of polystyrene spheres (3 – 8 mm sizes), agar gel, and NaN_3 as antiseptic simulating multi-textured lesions were embedded in the phantom.

[17] Using Laplacian of Gaussian, Wavelets, Exponential, Logarithm, Square, Square Root and 2D Local Binary Pattern

[18] Three green kiwis kept in ultrasound gel within a tough plastic box of suitable size. Intensity normalized using piecewise linear histogram matching. Two spatial resolution (1mm^3 and 2mm^3)

[19] The phantom consists of two modules—TMR009 (top) and TMR007 (bottom)—each containing over 100 spherical fiducials and solid test components filled with an MRI signal-generating liquid (96.4% distilled water, 2.5% PVP, 0.9% sodium chloride, and <0.2% each of potassium sorbate, copper sulfate, and blue food coloring; all by weight). QA cylindrical phantom filled with distilled water, featuring one central and four surrounding cavities for ionization chamber insertion

[20] Four filtered and transformed images including Laws, wavelets, Laplacian of Gaussian, and fractal analysis.

[21] Acrylic plastic cylinder with an inside diameter of 190 mm and an inside length of 148 mm.

[22] Homogenous phantom made of eight 30-ml tubes filled with demineralized water mixed with increasing gadolinium chelate concentration: 0.25 – 2 mmol/l and a central demineralized water tube. Heterogenous phantom (Eurospin phantom made of glass cylinder, filled with 1% copper sulphate) where two central columns had six 30-ml tubes filled with polystyrene beads of different diameters (1, 2 and 3 mm) in an agarose gel solution (2%), either pure or mixed with 0.25 mmol/l gadolinium chelate, and two 30-ml tubes filled with agarose gel solution (2%) and two different concentration of gadolinium chelate.

[23] Composed of polyethylene tubes (inner diameter of 28 mm) filled with different diameters of polystyrene spheres and an agar gel solution.

[24] Composed of liver (with six lesions inside), portal vein, two kidneys, lung, abdominal aorta, vena cava, vertebrae, and six ribs. It had volume of 26 ×19 × 12.5 cm^3

[25] It is elastomer membrane simulating the skin and subcutaneous fat layer of breast in patients and are filled with a branded gel (Zerdine). 5 - 10 cystic lesions (5–10 mm) and 10 - 15 dense lesions (5–10 mm) are included in the gel.

[26] Container filled with 0.4mM $MnCl_2$ solution as background and four inserts of polystyrene spheres of different diameters (1 to 8 mm) and 0.1% water solution of agar were embedded in the solution.

[27] Using Laplacian of Gaussian—LoG, wavelet, square, square root, logarithm, and exponential

[28] Named T1 Mapping and Extracellular volume Standardization (T1MES) initially designed for T1 mapping accuracy measurement of cardiac MRI. The phantom scanned where TE varied from 80 to 110 ms with TR of 4000 ms, while the TR varied from 3000 to 5000 ms with TE of 100 ms.

[29] Using wavelet, Laplacian of gaussian (LoG), square, square root, logarithm, exponential

[30] Using different scanner types, inversion recovery times, flip angles, and image preprocessing (bin discretization, Wavelet filter, Laplacian of Gaussian, logarithm, square, square root, and gradient filters)

[31] Four 6 mL test tubes filled with different materials to create varying image textures. Tube 1 contained vitamin E pills; Tube 2, chopped plastic IV tubing and $CuSO_4$ doped water; Tube 3, rolled medical gauze and $CuSO_4$ doped water; and Tube 4, 6 cm capillary tubes filled with $CuSO_4$ doped water.

[32] Three methods to quantize the original image intensity into 32, 64, 128, and 256 number of grey level intensity bins. This includes Lloyd-Max quantization, uniform quantization, and equal-probability quantization methods.

[33] Phantom consisted of 10 VOI including liver with six lesions inside, the portal vein, two kidneys, partial lung, the abdominal aorta, the vena cava, a vertebra, and six ribs, to mimic human abdomen

[34] Includes TR, TE, ETL, phase encoding direction, turbo direction, number of start-up RF pulse, RF pulse train flip angle pattern, driven equilibrium, and partial Fourier factor

[35] Transformed using LoG with the sigma levels of 1and 3, 1 level of wavelet decompositions yielding 8 derived images and the images derived using Square, Square Root, Logarithm and Exponential filters.

[36] Homogenous phantom made of eight 30-ml tubes filled with demineralized water mixed with increasing gadolinium chelate concentration: 0.25 – 2 mmol/l and a central demineralized water tube. Heterogenous phantom (Eurospin phantom made of glass cylinder, filled with 1% copper sulphate) where two central columns had six 30-ml tubes filled with polystyrene beads of different diameters (1, 2 and 3 mm) in an agarose gel solution (2%), either pure or mixed with 0.25 mmol/l gadolinium chelate, and two 30-ml tubes filled with agarose gel solution (2%) and two different concentration of gadolinium chelate.